

AWS Managed AI Services

Comprehend, Polly, Transcribe, Rekognition & Lex

AWS Machine Learning Associate • Part 14

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What We'll Cover



What are managed AI services?



The five services at a glance



Comprehend & Polly



Transcribe & Rekognition



Lex & combining the services



Managed AI vs. SageMaker

What Are Managed AI Services?



Intelligence as an API

- **Pre-trained models run by AWS**
  Used through simple API calls
- **No ML expertise needed**
  Nothing to train or maintain
- **Scales automatically**
  Pay only for what you use

Comprehend | **Polly** | **Transcribe** | **Rekognition** | **Lex**

Ideal for common tasks: understanding text, recognizing speech, analyzing images.

The Five Services at a Glance



Comprehend

Text → Insights

Sentiment, entities, key phrases



Polly

Text → Speech

Lifelike spoken audio



Transcribe

Speech → Text

Transcripts & subtitles



Rekognition

Image / Video → Labels

Objects, faces, text, moderation



Lex

Conversation → Bot

Chatbots & voice bots

Amazon Comprehend: Understanding Text



Sentiment Analysis

Positive, negative, neutral or mixed. Great for reviews & support tickets



Entity Recognition

Finds people, places, organizations & dates



Key Phrases & Language

Extracts key phrases and detects the language



PII Detection

Finds personal information, like names & addresses, so you can redact it

Custom classification lets you train it on your own labeled categories.

Amazon Polly: Text to Speech



Lifelike Voices

Including neural voices that sound very natural



Many Languages

A wide choice of voices across many languages



SSML Control

Markup for pauses, emphasis, speed & pronunciation



Lexicons

Define how specific words are said, like product names or acronyms

Common uses: e-learning narration, accessibility, voice in apps, reading articles aloud.

Amazon Transcribe: Speech to Text

Batch & Streaming

Transcribe recorded files, or live audio as it happens

Speaker Identification

Tells different speakers apart in a conversation

Custom Vocabulary

Teach it domain-specific terms and names

Language ID & Redaction

Detects the language, redacts sensitive info, generates subtitles

Common uses: call center transcripts, meeting notes, captioning video.

Amazon Rekognition: Images & Video



Objects & Scenes

Detects things like a dog, a beach or a car



Facial Analysis

Detect faces, compare two faces, search a face collection



Text in Images

Reads text that appears inside images



Content Moderation

Flags inappropriate or unsafe content

Custom Labels lets you train it to spot your own objects, like your products.

Amazon Lex: Chatbots & Voice Bots



Intent

What the user wants, like booking a hotel



Utterances

The different phrases people use to ask for it



Slots

The details Lex must collect, like the city and dates



Fulfillment

The action that runs once it has everything, often an AWS Lambda function

The same technology that powers Alexa, with speech recognition and language understanding built in.

Combining the Services: Customer Support

1



Transcribe

Turns call recordings into text

2



Comprehend

Finds sentiment and the key issues

3



Lex

Runs a chatbot for common questions

4



Polly

Speaks the answers aloud

Each step is just an API call, so mix and match to fit your use case.

Managed AI vs. SageMaker

A standard task: sentiment, speech or image labels	**Managed AI Services**
You want results fast, without ML expertise	**Managed AI Services**
You need a custom model with full control	**Amazon SageMaker**
Tailor a managed service with your own data	**Custom options**

Recap

- Managed AI services: pre-trained intelligence through simple APIs.
- Comprehend understands, Polly speaks, Transcribe listens, Rekognition sees, Lex converses.
- Reach for SageMaker when you need something fully custom.

Thanks for watching!